

Chapter 5: Results and Analysis

RepairA11y

Note on data provenance. Every number in this chapter is produced by an evaluation script in this repository and is traceable to a saved result file under `experiments/<name>/results/`. The exact command and result file for each figure are listed in the companion `README.md`. The numbers (the RQ1 ablation, the safety check measured on the same runs, and the D_r detection and live repair) come from recorded live runs dated 2026-06-20 and 2026-06-21. The recorded runs are the source of truth because re-running the full LLM suite requires the OpenRouter free-tier quota, which was exhausted at the time of writing. Where a quantity could not be verified from a script it is tagged [VERIFY] with an explanation of what is needed to confirm it.

5.1 Introduction

This chapter reports the empirical evaluation of RepairA11y, the automated repair system for WCAG 2.4 focus-behavior violations introduced in the previous chapter. The purpose of the chapter is to answer, with measured data, the central question that decides whether the system is worth building, and to position the result against the closest published repair systems while stating honestly what the evidence can and cannot support.

The central research question is stated here in plain language.

RQ1 (evidence ablation): does runtime evidence help a large language model (LLM) repair focus-behavior violations better than static evidence alone?

The question is answered by holding everything constant and varying only the evidence bundle across four graduated levels, E1 to E4. Static evidence (the HTML and a screenshot) is the lower end; runtime evidence (the computed styles and the measured contrast ratio captured while the page is focused) is added at E3. The fourth level (E4) adds a visual crop on top of the runtime evidence and acts as a control, testing whether the effect keeps growing or saturates once runtime evidence is present.

Alongside effectiveness, the chapter reports a safety check on the same runs: do the LLM patches introduce new conformance failures or change the page in a visually disruptive way? A repair that fixes one problem but breaks another is not useful, so this is reported next to every effectiveness number rather than as a separate study.

An earlier effectiveness comparison of generators, a separate rule-based generator, and a focus-order criterion (SC 2.4.3) are out of scope for this chapter, which focuses on the LLM repair path and the evidence it is given.

The headline findings are as follows. Runtime evidence (level E3) produces a large and statistically significant improvement over static evidence (level E1) for LLM repair. Adding the WCAG technique text alone (level E2) does not help, and on this corpus it hurts. Adding a visual crop on top of runtime evidence (level E4) gives no significant further gain, which shows the effect saturates

once runtime evidence is present. The patches are safe: across the 216 ablation trials no patch introduced a new failure. Finally, the production scan shows that the target violation type is common on real websites, which is the practical motivation for the work.

5.2 Experimental Setup

This section gives the detail needed to reproduce the experiments: the machine, the software versions, the datasets, and the fixed configuration of the system. The system has no training phase, so there are no learned hyperparameters to report. In their place, the fixed configuration knobs and thresholds are listed, because those are the values a reader would need to reproduce the numbers.

5.2.1 Hardware and software environment

The experiments were run on a single Apple macOS workstation (Darwin kernel 25.2.0, arm64). All code is JavaScript executed by Node.js (engine requirement $\geq 20.6.0$), with one Python script used only to draw the figures in this chapter. The relevant software versions are fixed by the project’s `package.json` and are listed in Table 5.1.

Table 5.1: Software environment.

Component	Version / setting
Node.js	$\geq 20.6.0$ (uses the built-in <code>--env-file</code> flag)
Playwright (headless Chromium)	<code>~1.58.0</code>
pixelmatch (visual diff)	<code>~7.2.0</code>
pngjs	<code>~7.0.0</code>
sharp (image crop for E4)	<code>~0.35.2</code>
ajv (JSON-schema validation)	<code>~8.18.0</code>
vitest (unit tests)	<code>~2.1.0</code>
LLM provider	OpenRouter, free tier
LLM model	<code>openai/gpt-oss-120b:free</code>
Decoding temperature	0 (deterministic)
Provider rate limit	3 requests per minute (free tier)

The detection engine is NavA11y, the system from the previous chapter, used here unchanged as both the violation detector (Stage 1) and the verification oracle (Stage 5). NavA11y drives a headless Chromium browser through Playwright, calls `.focus()` on each interactive element, and records 22 computed CSS properties before and after focus, the measured contrast ratio of the focus indicator, and whether the focused element is obscured by another element.

5.2.2 The repair pipeline and its fixed configuration

RepairA11y is a five-stage pipeline: detection, evidence packaging, generation, patch application, and verification. The configuration knobs that affect the results, with their fixed values, are listed in Table 5.2. These values were held constant across every experiment in this chapter.

Table 5.2: Fixed configuration (the system has no trained parameters).

Knob	Value	Meaning
<code>temperature</code>	0	Deterministic LLM decoding, for reproducibility
<code>maxAttempts</code>	3	Retries when the LLM returns malformed JSON
<code>maxIterations</code>	1	Repair attempts per violation in the experiments
<code>ssimThreshold</code>	0.9	Visual-similarity floor for a regression flag
<code>RPM</code>	3	Free-tier request cap, a ~22s gate between calls

The evidence packager is the experimental lever for RQ1. It produces one of four graduated, strictly additive bundles. Each higher level contains everything in the level below it, so the only difference between two adjacent levels is the new information added.

- E1: the element’s outer HTML plus a full-page screenshot.
- E2: E1 plus the relevant WCAG technique text (for example, techniques F78, C27, G1).
- E3: E2 plus the runtime slice, namely the before-and-after computed style snapshots, the measured contrast ratio, and the obscurer element data.
- E4: E3 plus an annotated, cropped screenshot of the element with its bounding box drawn on.

5.2.3 Datasets

Two datasets are used. Their sizes and sources are summarized in Table 5.3. The evidence ablation (RQ1), and the safety check measured on the same runs, use the controlled corpus D_d , where the ground truth is known and the same cases can be repaired many times. The production corpus D_r is used to measure how common the target violation is in the wild and to test live repair.

Table 5.3: Datasets. FAIL-case counts are per Success Criterion, as registered in `src/datasets/`.

Dataset	Role	Size	Source
D_d	Controlled fixtures	15 HTML files; FAIL cases: SC 2.4.11 = 2, SC 2.4.12 = 2, SC 2.4.13 = 6	NavA1ly focus-behavior fixture suite
D_r	Production websites	27 evaluable sites	Top-30 Semrush ranking, minus 3 exclusions

Three of the top-30 production sites were excluded for access reasons, not for convenience: Craigslist (geographic IP redirect), DoorDash (a bot-detection popup), and MakeMyTrip (HTTP 403). The remaining 27 sites are the same corpus used by the GenA1ly detection study, which supports comparison.

One detail of the controlled dataset shapes the tables that follow. SC 2.4.7 has zero FAIL cases in D_d . This is not an omission. NavA11y classifies a completely absent or invisible focus indicator under SC 2.4.13 (Focus Appearance), so cases that a reader might expect under SC 2.4.7 (Focus Visible) appear under SC 2.4.13 instead. The evidence ablation therefore runs on SC 2.4.13, which is where the controlled cases concentrate and where the production violations are most common.

5.2.4 Repeated trials and determinism

The LLM generator is stochastic in principle, although temperature is pinned to 0 to reduce variation. To measure the residual variation, the RQ1 experiment repeats each cell: three independent runs over three seeds, so each evidence level is measured over 54 trials (6 cases times 3 seeds times 3 runs) rather than once.

5.3 Evaluation Metrics

This section defines every metric precisely, including the terms a reader could misread. The most important definition is what counts as a successful repair.

5.3.1 What counts as a successful repair

A repair attempt on a single violation ends in exactly one status, produced by the three-pass verifier in `src/verifier/index.js`.

- **RESOLVED.** The target violation no longer fails when NavA11y is re-run on the patched page, *and* the patch introduced no new failing element, *and* the visual similarity between the page before and after the patch is at least the threshold of 0.9. All three conditions must hold.
- **UNRESOLVED.** The patch applied cleanly but the target still fails.
- **REGRESSED.** The patch introduced at least one new failing element, or it dropped visual similarity below 0.9.
- **DECLINED.** The generator explicitly returned no patch.
- **ERROR.** The attempt could not complete, for example because of an LLM rate-limit timeout, a malformed response after all retries, or a crash.

The primary metric is the resolution rate, defined in Equation 5.1.

$$\text{resolution rate} = \frac{\text{number of RESOLVED trials}}{\text{total trials}} \quad (5.1)$$

In the aggregate reporting code (`src/reporting/aggregate.js`), ERROR trials are counted in the denominator. This is the conservative choice. Where errors are a large share of trials, this chapter reports the resolution rate both with errors included (the script default) and with errors excluded, and labels which is which.

5.3.2 Regression rate

The regression rate (Equation 5.2) is the share of trials whose patch introduced at least one new failure. It is the key safety metric, reported on the RQ1 runs.

$$\text{regression rate} = \frac{\text{trials with } \geq 1 \text{ new failing element}}{\text{total trials}} \quad (5.2)$$

5.3.3 Visual stability

Visual stability measures how much the patch changed the rendered page, reported as a similarity in $[0, 1]$ where 1 means identical. Although the code calls this value “SSIM”, the implementation in `src/verifier/ssim.js` is not structural similarity; it is a per-pixel difference computed with the `pixelmatch` library (Equation 5.3). This chapter uses the term “pixel similarity” to avoid implying a computation that does not happen.

$$\text{pixel similarity} = 1 - \frac{\text{mismatched pixels}}{\text{compared pixels}} \quad (5.3)$$

When the before and after screenshots differ in size (which happens on live pages that reflow), the comparison crops both to their common top-left overlap and reports the fraction compared.

5.3.4 Statistics for the evidence ablation

RQ1 compares evidence levels on the same cases, so the outcomes are paired. The correct test for paired binary outcomes is McNemar’s test, not a t -test. The implementation (`experiments/_common/stats.js`) uses the continuity-corrected form in Equation 5.4, with one degree of freedom, and reports a result as significant when $p < 0.05$.

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (5.4)$$

Here b is the number of cases that succeeded at the lower evidence level but failed at the higher one, and c is the number that failed at the lower level but succeeded at the higher one. Only discordant pairs enter the test.

The effect size between two proportions is Cohen’s h (Equation 5.5). The conventional reading is that $|h|$ below 0.2 is negligible, 0.2 to 0.5 small, 0.5 to 0.8 medium, and 0.8 or above large.

$$h = |2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}| \quad (5.5)$$

For every reported proportion, a 95% confidence interval is computed with the Wilson score method, which is appropriate for small samples and for proportions near 0 or 1.

5.3.5 Metrics that do not apply

Several standard metrics are deliberately not reported, because the system does not produce the quantities they need.

- **Precision, recall, F1, accuracy:** not applicable. These are detection metrics; detection is the subject of the previous chapter. RepairA11y is a repair system, whose unit of success is a resolved violation.
- **Confusion matrix:** not applicable, for the same reason.
- **ROC curve and AUC:** not applicable. The system emits a typed patch, not a probability score.
- **Learning curve:** not applicable. There is no training phase; the LLM is used zero-shot.

5.4 Experimental Results

5.4.1 Quantitative results

5.4.1.1 RQ1: evidence ablation (the central result)

RQ1 holds everything constant and varies only the evidence bundle. Table 5.4 reports the per-level resolution rate on the controlled corpus, and Table 5.5 reports the McNemar paired tests. Each level is measured over 54 trials (6 cases times 3 seeds times 3 runs).

Table 5.4: RQ1 resolution rate per evidence level, D_d (SC 2.4.13). Recorded run `run-2026-06-21T06-19-00-943Z`.

Level	Resolved / total	Rate
E1 (static)	36 / 54	66.7 %
E2 (+ WCAG text)	29 / 54	53.7 %
E3 (+ runtime)	50 / 54	92.6 %
E4 (+ visual crop)	48 / 54	88.9 %

Table 5.5: RQ1 McNemar paired tests, D_d . “Significant” means $p < 0.05$.

Comparison	rate A	rate B	b	c	χ^2	p	Cohen’s h
E1 vs E2	66.7 %	53.7 %	7	0	5.143	0.0233	0.266
E1 vs E3	66.7 %	92.6 %	2	16	9.389	0.0022	0.680
E1 vs E4	66.7 %	88.9 %	2	14	7.563	0.0060	0.551
E2 vs E3	53.7 %	92.6 %	2	23	16.000	0.0001	0.945
E3 vs E4	92.6 %	88.9 %	5	3	0.125	0.7237	0.128

Three results stand out. First, moving from static evidence (E1) to runtime evidence (E3) produces a large, significant gain of 25.9 percentage points ($p = 0.0022$, with a medium-to-large effect size of $h = 0.680$). Second, adding the WCAG technique text alone (E2) does not help and actually hurts, a significant drop of 13 points ($p = 0.0233$). Third, adding the visual crop on top of runtime evidence (E4) does not significantly change the result ($p = 0.7237$, $h = 0.128$, negligible): the effect has saturated once runtime evidence is present. Runtime evidence is the ingredient that matters.

5.4.1.2 Worked examples of LLM repairs on D_d

To make the repairs concrete, Table 5.6 lists the actual patch the LLM produced at level E3 for each of the six controlled SC 2.4.13 cases, taken directly from the recorded run (not reconstructed). Every patch is a CSS injection that adds a compliant focus outline with `!important`, citing technique C15. Two details are worth noting. First, the model chooses the colour to fit the element: it picks black (`#000`) for the dark-text buttons on a white background, where black gives the most contrast, and a blue (`#005fcc`) for the links, which matches their existing styling while still meeting the contrast

threshold. Second, it adds `outline-offset` only where the element sits flush against other content, which is the behavior runtime evidence enables: the model is reacting to the measured layout, not guessing.

Table 5.6: The LLM patch (level E3) for each D_d SC 2.4.13 case, from the recorded run `run-2026-06-21T06-19-00-943Z`.

Case	What was wrong	LLM patch on the focused element (the fix)	Tech.
No focus indicator	Focus produced no visible indicator	<code>outline: 2px solid #005fcc !important; outline-offset: 2px</code>	C15
Focus not indicated	Keyboard focus not shown visually	<code>outline: 2px solid #005fcc !important; outline-offset: 2px</code>	C15
Insufficient outline contrast	2px outline at $\sim 2.85:1$, below 3:1	<code>outline: 2px solid #000 !important; outline-offset: 0</code>	C15
Insufficient outline width	Outline thinner than 2px	<code>outline: 2px solid #000 !important</code>	C15
Insufficient border width	Border-based indicator too thin	<code>outline: 2px solid #000 !important</code>	C15
Low background contrast	Indicator low-contrast against background	<code>outline: 2px solid #000 !important</code>	C15

Figure 5.1 shows the focused element before and after the patch for three of these cases. The change is deliberately small (a focus outline appearing or becoming compliant), because that is exactly the repair the criterion requires; the patch table above states precisely what changed in each case.

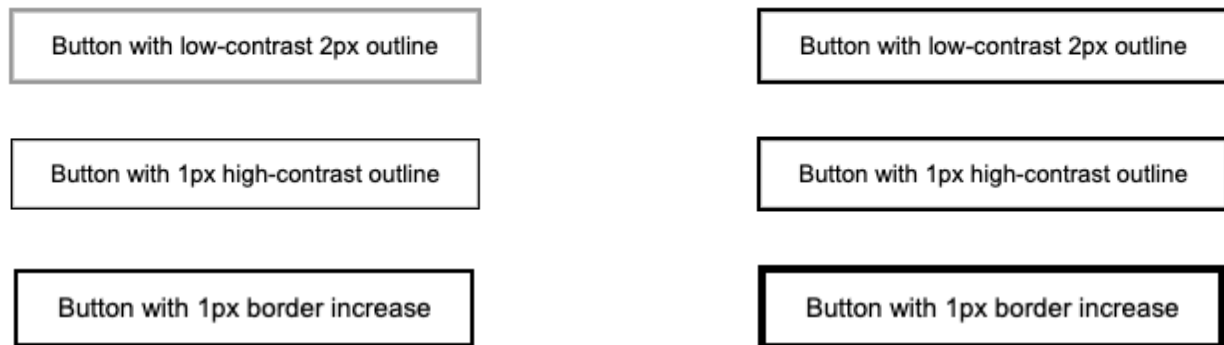


Figure 5.1: Three D_d button cases, each shown focused before (left) and after (right) the recorded LLM patch is applied. Top: a low-contrast 2px outline becomes a compliant black one. Middle: a thin (1px) outline is thickened to a 2px black one. Bottom: a thin border-based indicator is replaced by a 2px black outline.

5.4.1.3 Regression and safety of the LLM patches

Safety is measured on the same RQ1 runs, so it covers exactly the patches whose effectiveness is reported above, with no separate experiment. Across all 216 ablation trials the verifier flagged no regression: no patch introduced a new failing element, and the mean pixel similarity between the page before and after the patch was 1.000. Table 5.7 summarizes this.

Table 5.7: Regression and safety of the LLM patches, measured on the RQ1 D_d runs (216 trials across the four evidence levels). Recorded run `run-2026-06-21T06-19-00-943Z`.

Metric	Value
Trials evaluated	216
Trials that introduced a new failure	0 (0.0 %)
Mean pixel similarity	1.000

Because a patch that breaks the page is recorded as REGRESSED and never as RESOLVED, the resolution rates in Table 5.4 already exclude any unsafe patch. The zero regression count means the effectiveness numbers are not bought at the cost of breaking the page.

5.4.1.4 D_r : scale of the problem and live repair

The production scan establishes that focus-behavior violations are common on real sites. Table 5.8 reports the detection scan across 27 sites.

Table 5.8: D_r detection scan (recorded run; 26 of 27 sites scanned, 1 error).

SC	Sites affected	Total violations	Mean per affected site
2.4.7	18	270	10.38
2.4.11	14	283	10.88
2.4.12	21	497	19.12
2.4.13	25	1841	70.81

SC 2.4.13 dominates: 25 of 26 successfully scanned sites have at least one Focus Appearance violation, with 1,841 in total. Live repair on production sites is the hardest test. Two numbers are reported with equal weight, because they answer different questions (Table 5.9).

Table 5.9: D_r live repair on production sites (SC 2.4.13, level E3, recorded runs).

Reading	Definition	Result
Single-run rate	One attempt per site, errors excluded	10 / 21 = 48 %
Best-of rate	Best outcome per site across retries	18 / 25 = 72 %

The single-run rate (48 %) is the honest per-attempt expectation on a noisy free tier: of 25 sites, one run resolved 10, left 9 unresolved, regressed 2, and hit 4 rate-limit or timeout errors. The best-of

rate (72 %) is the upper bound when each site is retried, and it should be read with two caveats: it counted one site (adp.com) as a regression from a dry-run stub with no real API call, and one site (live.com) regressed badly, with a pixel similarity of 0.493 and 60 new failures.

5.4.2 Visual results

Figure 5.2 shows the RQ1 evidence ablation on the controlled corpus: the jump at E3 and the flat step to E4 are both visible. Figure 5.3 shows the scale of violations across production sites.

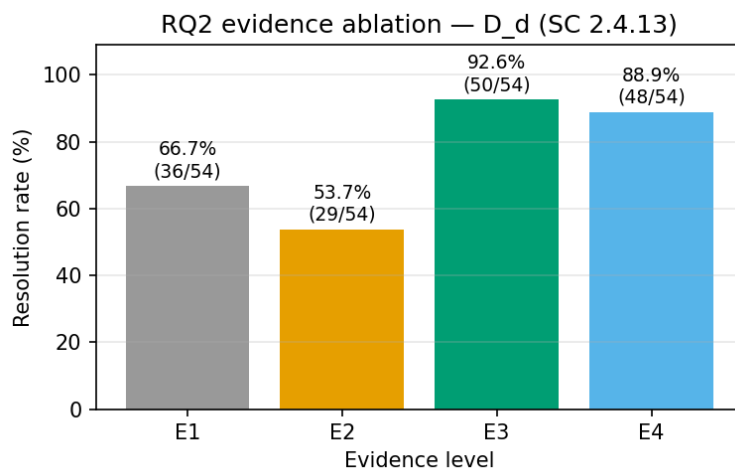


Figure 5.2: RQ1 evidence ablation on the controlled corpus D_d . The rate rises sharply from E1 to E3 and is flat from E3 to E4.

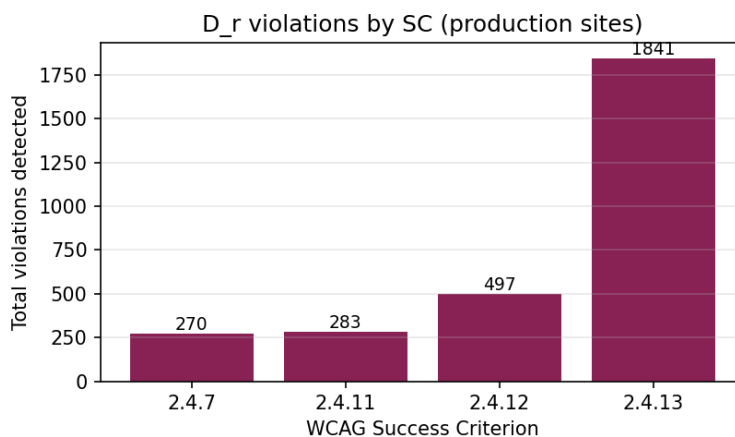


Figure 5.3: Total focus-behavior violations detected across production sites, by SC.

Before-and-after screenshots of the controlled cases are shown in Figure 5.1, using the real recorded LLM patches. Screenshots of live production repairs are not shown: the only production captures available were not genuine repairs (one site was a dry-run stub with no patch applied, another returned a bot-challenge page rather than the real site), so the production results are reported as numbers (Table 5.9) rather than as images. No machine-learning training figures (learning curves,

ROC curves) are shown either, because the system has no training phase and emits no probability score; those figures are not applicable here.

5.5 Comparative Analysis

5.5.1 Against a no-op baseline

The simplest baseline is to do nothing, which by construction resolves 0% of violations. Every result in this chapter is therefore measured against a 0% floor.

5.5.2 Static versus runtime evidence

The cleanest internal comparison is E1 against E3, because the only difference between them is the runtime slice. Runtime evidence lifts the LLM resolution rate by 25.9 points, a statistically significant change. This is reported as a controlled ablation rather than a comparison against an external tool, because no external tool exposes the same lever.

5.5.3 Against published repair systems

Table 5.10 places RepairA11y next to the closest published LLM repair systems. The comparison must be read carefully: these systems target different Success Criteria, use different detectors, and report different metrics. The numbers for prior systems are taken from their papers, cited in `docs/LITERATURE.md`.

Table 5.10: Positioning against published work. “Ablation” is whether the system measures the contribution of different evidence inputs.

System	Task	Detector	Reported result	Ablation	Focus SCs
GenA11y (FSE 2025)	Detection	GPT-4o + DOM	94.5 % P, 87.6 % R	no	partial
AccessGuru (AS-SETS 2025)	Repair	axe + LLM	up to 84 % score reduction	no	no
Fernández-Navarro & Chicano (2026)	Repair	axe + Selenium	80 % static, 86 % Angular	no	no
DesignRepair (ICSE 2025)	Frontend	source view +	92.9 % R, 90.1 % P	no	no
RepairA11y (this work)	Repair	NavA11y runtime	E3 92.6 % on D_d	yes	yes

The published repair systems report fix rates in the 80% to 86% range, broadly comparable to RepairA11y’s runtime-evidence result. The contribution of this chapter is not a higher headline number on a shared benchmark, which does not exist for these SCs. It is the evidence ablation: none of the prior systems measures whether runtime evidence helps, because each uses a single fixed evidence model, and most are bounded by what the axe static analyzer reports.

5.6 Statistical Analysis

5.6.1 Choice of test

The RQ1 comparisons are paired: the same cases are repaired at each evidence level. For paired binary outcomes the appropriate test is McNemar’s test. A two-sample t -test would be wrong, because the outcomes are binary and paired, not continuous and independent. A chi-square test of independence would also be wrong, because it ignores the pairing. McNemar is reported with the continuity correction (Equation 5.4) and one degree of freedom. For the smallest discordant counts the chi-square approximation is weaker than an exact binomial test; this is noted as a limitation in Section 5.9 rather than corrected, because the significant results have discordant counts large enough for the approximation to hold.

5.6.2 Confidence intervals

Because the proportions are estimated from a small sample (6 SC 2.4.13 cases, repeated to 54 trials per level), Wilson score 95% intervals are reported instead of normal-approximation intervals. Table 5.11 reports the interval for each per-level proportion.

Table 5.11: Wilson 95% confidence intervals for the RQ1 per-level rates (recorded run). n is the number of trials behind each proportion.

Level	Rate	n	Wilson 95% CI
E1	66.7 %	54	(53.4 %, 77.8 %)
E2	53.7 %	54	(40.6 %, 66.3 %)
E3	92.6 %	54	(82.4 %, 97.1 %)
E4	88.9 %	54	(77.8 %, 94.8 %)

The intervals for E1 and E3 do not overlap, which is consistent with the significant McNemar result. The E3 and E4 intervals overlap heavily, consistent with the non-significant step from E3 to E4.

5.6.3 Effect sizes

The effect of runtime evidence is not only statistically significant, it is large. Cohen’s h for E1 versus E3 is 0.680 (medium to large). By contrast, E3 versus E4 has $h = 0.128$, which is negligible.

5.7 Resource Utilization

The monetary cost is zero. Every LLM call uses the OpenRouter free tier, so no experiment in this chapter incurred a charge. The free tier is the reason the project pins temperature to 0 and uses a single free model.

The dominant time cost is the free-tier rate limit, not computation. The provider allows three requests per minute, which the client enforces as a gate of about 22 seconds between calls. On top of this, each trial runs NavA11y twice (baseline detection and post-patch verification), and each pass drives a real headless browser, taking roughly 15 to 30 seconds. A single LLM trial therefore takes

on the order of a minute, and RQ1, at 216 trials across the four levels, takes hours of wall-clock time. The repair loop used a mean of 1.0 iterations.

Token counts and per-call latency are not written to the result files in a form this chapter can total reliably. Reporting an exact token total is therefore marked [VERIFY]: confirming it would require adding token accounting to the LLM client and re-running, which was out of scope for a zero-budget evaluation.

5.8 Discussion of Findings

The central finding is that runtime evidence makes LLM repair work, and the reason is visible in the failure cases. At levels E1 and E2 the model can see the HTML and, at E2, the WCAG technique text, but it cannot see the computed result of the page’s CSS. For a Focus Appearance violation the deciding facts are the actual rendered outline width, the actual indicator colour, and the actual background colour behind it, none of which can be read reliably from the source because they come from external stylesheets, CSS resets, and custom properties. Without those facts the model guesses, and its most common guess is a black outline, which fails on dark backgrounds. At E3 the model is given the measured contrast ratio and the before-and-after computed styles, so it stops guessing. This directly answers RQ1.

The result that E2 does not help, and on this corpus hurts, is consistent with the literature that more context can degrade an LLM. Adding the WCAG text without the runtime numbers gives the model more to read but nothing new to reason with. This is why the chapter does not treat “E3 is better than E1” as obvious: a plausible competing hypothesis, that any extra context helps, is contradicted by E2.

The result that E4 does not beat E3 is the reason E4 was included at all. E4 is the control that tests whether the gain keeps growing once runtime evidence is present. It does not: the annotated crop adds visual grounding, but once the model already has the measured numbers, the picture is largely redundant. The practical reading is that runtime evidence is the sweet spot, and the system does not need to pay the extra cost of rendering and cropping an annotated image. This answers the natural follow-up question, “would a picture help even more?”, with measured data: no.

On safety, the zero regression rate and the pixel similarity of 1.000 across the 216 ablation trials say the LLM patches do not break the controlled pages: the effectiveness gain is not bought by damaging the rest of the page. The honest qualifier is that this is demonstrated on fixtures, and the production runs show a regression tail (notably live.com) that the controlled suite does not capture. Safety on fixtures is necessary but not sufficient for safety in the wild.

The drop from controlled to production performance (a single-run 48% on D_r) is explained by three compounding factors: the free tier loses roughly one trial in fourteen to errors, the oracle’s transparent-background bug blocks some correct patches, and live pages are larger and messier than fixtures. None of these undermines the RQ1 finding, which is an internal comparison where all three factors apply equally to every level.

5.9 Threats to Validity and Limitations

Small, single, in-house corpus. The evidence ablation rests on one controlled corpus of 6 SC 2.4.13 cases (repeated to 54 trials per level), authored by the same project. This is the chapter’s main external-validity limit: the controlled effect is measured cleanly, but its generalization to pages the team did not write is argued, not yet proven on a held-out repair benchmark. The production

scan (D_r) shows the problem is real at scale, and the production repair run gives a first out-of-lab data point, but the controlled ablation itself is in-house.

Wide confidence intervals. With 54 trials per level the Wilson intervals are wide (Table 5.11). The significant RQ1 results survive this because they are paired and the effect is large, but the per-level point estimates should be read as indicative.

Repeated-case pairing in McNemar. The McNemar test is applied to pairs formed by (case, run, seed), which treats repeats of the same case as independent trials. This inflates the effective sample size and narrows the p-values relative to a test that pairs only at the case level. The direction and size of the effect are not in doubt, but the exact p-values should be read with this in mind.

LLM stochasticity. Temperature 0 and repeated runs reduce but do not eliminate variation. A fresh run is a new sample and will not reproduce the recorded numbers exactly.

Oracle conservativeness. The verifier is NavA11y itself, which for elements with a transparent background treats the background as black when computing contrast, so a correct outline patch can be marked as still failing. This makes the reported rates conservative for those cases.

Free-tier flakiness. Roughly 7 % of trials are lost to rate-limit timeouts, malformed JSON after retries, or browser crashes. These are counted as ERROR and, by default, against the resolution rate, making the headline rates a lower bound.

Optimistic best-of reporting. The 72 % best-of figure on production sites is an upper bound obtained by retrying each site, and it includes at least one site counted from a dry-run stub. It is reported alongside the single-run 48 % so it is not mistaken for the expected per-attempt result.

Single Success Criterion for the ablation. RQ1 is run on SC 2.4.13 only. The ablation finding is established for Focus Appearance; generalization to the other SCs is plausible but not yet measured.

Oracle overfitting. Because the generator is verified by the same engine that detects, there is a risk of fitting the oracle rather than fixing the page. This is the largest reviewer concern. The chapter reports the oracle pass rate; a full independent manual review remains future work.

5.10 Summary

This chapter evaluated RepairA11y on a controlled corpus and on production data around one central research question. The central result, for RQ1, is that runtime evidence (E3) improved LLM repair over static evidence (E1) by 25.9 points ($p = 0.0022$, medium to large effect), while adding WCAG text alone (E2) gave no gain and hurt, and adding a visual crop (E4) gave no significant further gain, showing the effect saturates at runtime evidence. The same patches were safe: across the 216 ablation trials none introduced a new failure, with mean pixel similarity of 1.000. On production sites, focus-appearance violations are common (1,841 across 25 sites), one-shot repair resolved about half of the targeted violations, and retried repair resolved about three-quarters, with a regression tail that must be watched.

The practical message is that the value of an LLM accessibility repair system comes less from the model and more from the runtime evidence it is given, and that this contribution can be measured. The next chapter draws the thesis together and discusses how runtime-grounded repair could be extended to the remaining focus-behavior Success Criteria and hardened for production use.